

CLIP-Free, Label-Free, Unsupervised Concept Bottleneck Models

U-F²-CBM

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What?

Concept Bottleneck Models currently:

- Rely on CLIP to extract concepts
- **Or** require manual image-concept annotations
- **AND** re-train a supervised classifier on top of the bottleneck layer

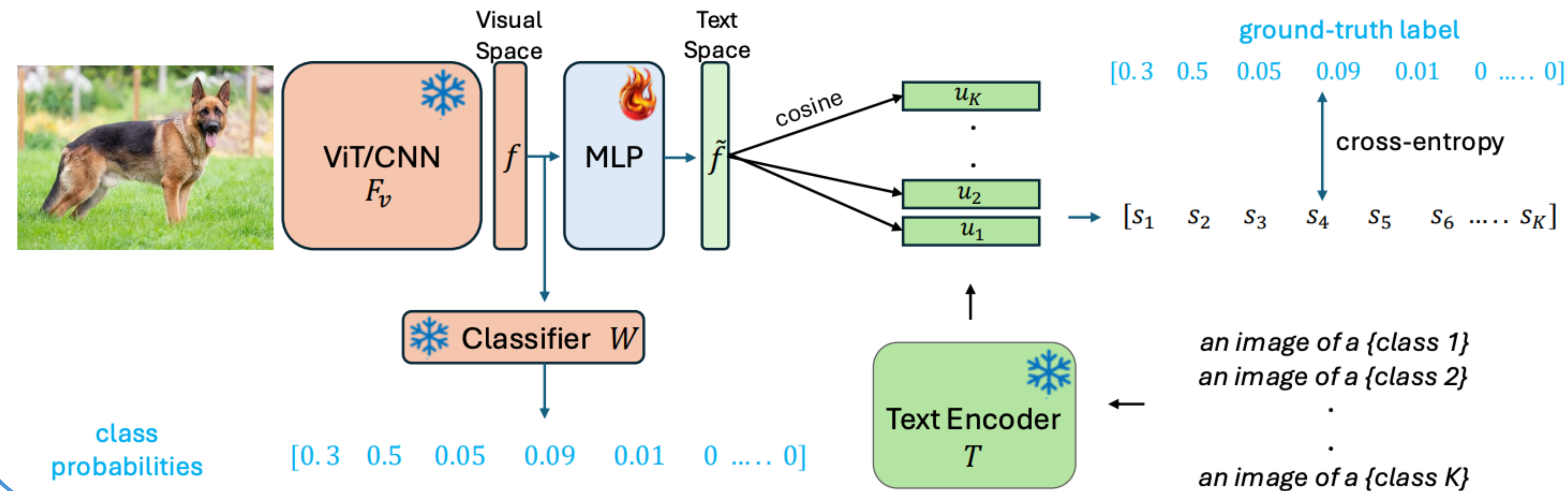
Our Solution

U-F²-CBM

Unsupervised • CLIP-Free • Label-Free

Fully Unsupervised Learning

Align the class-names distribution with the classifier's discrete distribution



Text-Unlock

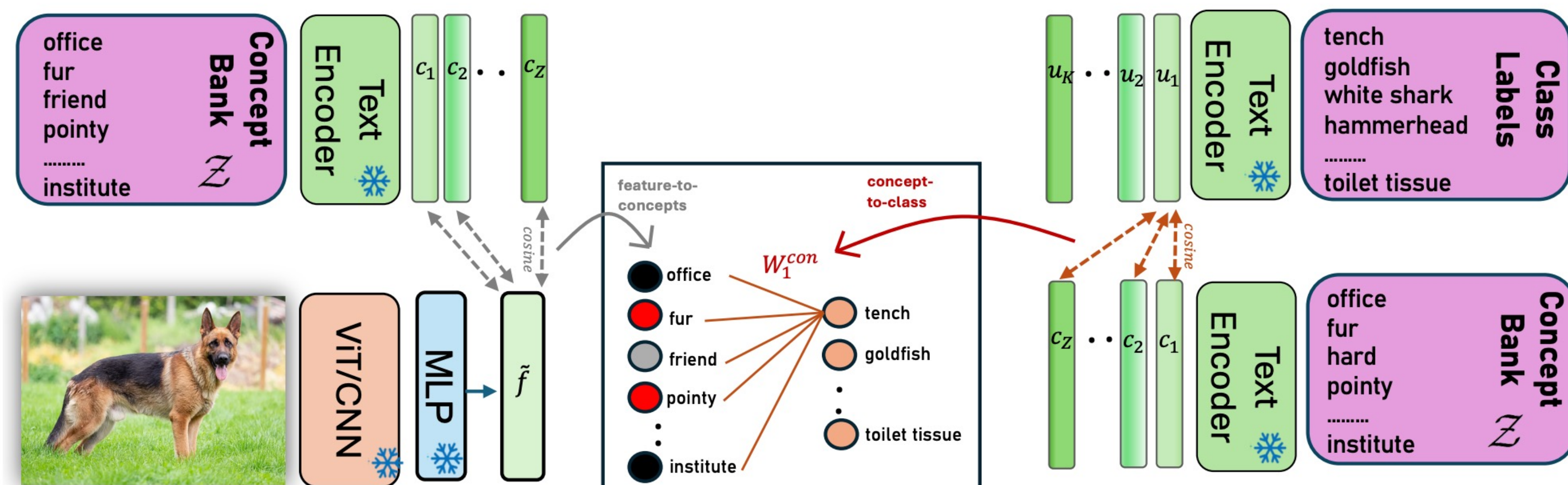
Training

Inference

U-F²-CBM

Concept Discovery

Unsupervised Derivation of the Linear Classifier



Summary of Results

(see paper for quantitative scores)

- 1 Text-Unlock maintains the classifier performance
- 2 U-F²-CBM achieves SoTA and beats CLIP-based CBMs
- 3 Beats CLIP-based approaches for Zero-Shot Image Captioning
- 4 CLIPifying any classifier with 400× less images

